

Arkadip Som

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EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur

Howrah, West Bengal, India

B.Tech in Civil Engineering

Aug 2024 – July 2028

- **CGPA:** 7.99 / 10.00 (Till 4th Semester)
- **Core Coursework:** Probability & Statistics, Linear Algebra, Geographical Information System, Numerical Methods

Begampur High School

Hooghly, West Bengal, India

Higher Secondary (Class XII) & Secondary (Class X)

January 2015 – April 2023

- **Higher Secondary Examination (WBCHSE) - 2023:** Secured **87.8%** aggregate; ranked **2nd** institutionally.
- **Secondary Examination (WBBSE) - 2021:** Secured **97.8%** aggregate; ranked **12th across the state of West Bengal** and **2nd** institutionally (merit evaluated via official board criteria).

PROFESSIONAL EXPERIENCE

Bureau of Indian Standards (BIS) — National Case Study Competition

Kolkata, India

Team Lead & Field Researcher

Jan 2026 – April 2026

- Led a **3-member team** selected nationally to evaluate industrial quality compliance, coordinating stakeholder interviews and site audits directly with BIS Kolkata leadership and UltraTech Cement Dankuni plant heads.
- Audited cement manufacturing workflows at UltraTech Cement plant, verifying live testing procedures against mandated Indian Standards (IS 269, IS 1489, IS 455); identified **1 critical fly-ash process deviation** and mapped **5 internal quality benchmarks** maintained beyond baseline code requirements.
- Synthesized complex plant test data and executive interviews into a **100-page** compliance report, surfacing **10 compliance framework gaps** and **15 actionable modernization recommendations** submitted for BIS review.

TECHNICAL PROJECTS

Climate Resilience & Strategic Capital Allocation Platform | [\[GitHub\]](#) [\[Live Link\]](#) Python, Scikit-Learn, Streamlit

Independent Developer

Jul 2026

- Engineered a composite Climate Resilience Score across major Indian metros using **Min-Max normalized indicators (heatwave, flood, water stress, population pressure, infrastructure readiness)** combined via hierarchical weighted formulas.
- Developed predictive regression models to analyze historical risk indices, isolating critical vulnerabilities across key threat sectors including **heatwave frequency, water stress, and urban flooding indicators**.
- Built a predictive layer using **OLS regression** on decadal climate baselines and compound population growth models to forecast **post-2026 vulnerability trends**, then designed a **rule-based engine** to dynamically route a user-adjustable interactive infrastructure pool (**up to 20,000 Cr**) across sectors by projected risk.

Climatic Analytics & Temperature Forecasting Engine | [\[GitHub\]](#)

Python, Pandas, Scikit-Learn

Independent Developer

Jan 2026

- Implemented an **end-to-end regression pipeline** using Scikit-Learn to model non-linear climatic patterns across a **3,190+ sample dataset**, isolating critical historical weather predictors.
- Engineered data preprocessing structures integrating seasonal **feature extraction, standard scaling** and **SelectKBest** feature selection to ensure model stability and prevent data leakage.
- Optimized a **RandomForestRegressor** model achieving a high-precision **0.981 test R^2 score** and an **RMSE of 0.171**, executing residual analyses to validate the complete absence of systematic bias.

LEADERSHIP AND RESPONSIBILITY

- **Publicity Team, INSTRUO 14.0 (Techno-Management Fest, IIST Shibpur)** — led outreach at Jadavpur University Salt Lake campus for 30+ events, with focused promotion of Hacksprint, the flagship hackathon (5L prize pool, 55% of the fest's 9L total)
- **Member, Society of Civil Engineers (SOCCE)** — co-organized technical seminars and departmental events

SKILLS

- **Programming & Databases:** Python, C, C++, SQL
- **Data & Analytics:** NumPy, Pandas, Matplotlib, Scikit-Learn, Regression Modeling, Statistical Analysis
- **Tools & Software:** MS Excel, PowerPoint, Git/GitHub
- **Languages:** English (Fluent), Bengali (Native), Hindi (Conversational)

CERTIFICATIONS & ACHIEVEMENTS

- GirlScript Summer of Code (GSSoC '26) — Open Source Contributor (Apr 2026)
- BCG Strategy Consulting & Introduction to Consulting Simulations — Forage (July 2026)